Practice Problems for Week 4

Operating and Systems Programming 2022/23

1. What is the purpose of threads?
2. Which of the following data is shared between threads?
   - Global variables
   - Data on stacks
   - Data in the heap
   - Open files
3. What is a critical section?
4. Why is it important that critical sections take as little time to execute as possible?
5. What may happen if a thread blocks during execution of a critical section?
6. Assume that `fun1` and `fun2` are executed by two different threads. Identify the critical sections in the following program fragment:

```
#include <stdio.h>
#include <stdlib.h>

int totalCalls = 0;

char *fun1 () {
    static int isExecuted = 0;
    char *result;

    isExecuted++;
    totalCalls++;
    result = malloc(5);
    return result;
}

char *fun2 () {
    static int isExecuted = 0;
```

```
    char *result;

    isExecuted++;
    totalCalls++;
    result = malloc(10);
    return result;
}
```

7. How would you protect the critical section in the above example? Can you use a read-write lock for this example?

8. Consider the following program fragment

   ```
       #include <stdio.h>
   #include <stdlib.h>

   char *msg = "This is a message";

   void fun1 () {
       if (msg != NULL) {
   msg = NULL;
       }
   }

   char *fun2 () {
       char *result;

       if (msg == NULL || (strlen(msg) > 5)) {
   return NULL;
       }
       result = malloc(6);
       strcpy(result, msg);
       return result;

   }
   ```

   where both functions may be executed as part of a multi-threaded system. Identify the critical sections and use appropriate locking to protect them. Your critical sections should be executed in as little time as possible.